

Honeypot-Driven Proactive Detection Network for Attacks in Smart Grids

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Abstract—This paper proposes a honeypot-driven active detection framework to address the growing cyber threats targeting smart grids. To address the limitations of existing detection methods and the lack of proactive defense capabilities, the framework integrates interpretable decision mechanisms with an adaptive modular architecture, through the dynamic weighting of the gated aggregation mechanism. It incorporates data preprocessing, honeypot interaction, dual-path detection, and decision fusion to more effectively capture diverse attack behaviors and strengthen overall detection reliability. The framework integrates a GBDT-based anomaly scorer with a dedicated Denial-of-Service (DoS) detector, and combines multidimensional evidence using a gated aggregation mechanism. Moreover, theoretical analyses establish a lower bound on feature distinguishability, prove model convergence, and demonstrate effective false-alarm suppression, providing formal guarantees for feasibility. Finally, experiments on real-world smart grid datasets show that the proposed framework accurately detects DoS, replay, and integrity attacks while maintaining strong overall performance. The results confirm its effectiveness as a theoretically grounded and proactive defense solution for smart grid security.

Note to Practitioners—Smart grids face unique cybersecurity challenges. Attacks like data replay or network floods can disrupt physical operations, leading to outages or equipment damage. Traditional IT security tools often struggle here, as they may only respond after the system has been attacked. This work addresses this gap by integrating deceptive honeypot nodes directly into the operational network architecture. These honeypots act as intelligent sensors, attracting and engaging attackers. The core innovation is a method to automatically convert the observed attacker interactions into clear, actionable signals. These signals are then fused with conventional traffic analysis in a decision framework to reliably distinguish between Replay attacks, Integrity tampering, Denial-of-Service attempts, and normal operations. For security teams, this means more precise alerts with contextual evidence, intercepting attacks before any damage occurs.

Index Terms—Smart grid security, honeypot interaction, active defense, attack detection.

Received 6 January 2026; revised 12 March 2026; accepted 2 April 2026. Date of publication 7 April 2026; date of current version 24 April 2026. This article was recommended for publication by Associate Editor M. Zhang and Editor Q. Zhao upon evaluation of the reviewers' comments. This work was supported in part by the National Natural Science Foundation (NNSF) of China under Grant 61890920 and Grant 08120003 and in part by the Fundamental Research Funds for the Central Universities under Grant DUT22RT(3)090. (*Corresponding author: Xi-Ming Sun.*)

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Digital Object Identifier 10.1109/TASE.2026.3681594

I. INTRODUCTION

EMERGING from the deep integration of information and communication technologies (ICT), the smart grid significantly enhances the efficiency, reliability, and sustainability of traditional power systems [1]. However, this transformation also introduces new cybersecurity challenges. The smart grid's distributed architecture, extensive sensor deployment, and deep integration with the Internet and IoT significantly expand its attack surface, exposing the system to a wide range of cyber threats, including false data injection [2], denial of service (DoS) attacks, data tampering, malware propagation, and eavesdropping, all of which can compromise its physical, communication, and control layers [3]. Attacks that target grid reliability are particularly critical. By compromising measurement devices (e.g. PMUs, SCADA), attackers can circumvent bad-data detection mechanisms [4], distort state estimation, and induce unsafe control decisions. Such failures in real-time monitoring may lead to frequency deviations, localized outages, or even large-scale system collapse.

To address these challenges, attack detection in smart grids is evolving toward deeper integration of data-driven AI methods, cross-layer coordinated defense, and improved real-time adaptability. Machine learning and deep learning are widely applied to attack identification: supervised and semi-supervised models achieve high accuracy in detecting false data injection attacks, while Convolutional Neural Networks (CNN)- and Long Short Term Memory(LSTM)-based architectures enhance time-series anomaly detection [5]. Generative adversarial networks(GANs) further enhance robustness against adversarial samples [6]. Hybrid detection frameworks that combine power system dynamics with data-driven methods are also gaining maturity, enabling cross-layer attack identification of coordinated attacks through residual analysis and state estimation deviation in Cyber-Physical Systems (CPS) [7]. Mitigation strategies such as adversarial training and moving target defense address model vulnerabilities and evolving attack environments, and online or reinforcement learning techniques enable real-time adaptation [8]. In addition, distributed detection architectures prevent single points of failure [9], and privacy-preserving techniques, including federated learning, support secure multi-party collaboration without exposing sensitive data [10].

Within this broader landscape, proactive defense has emerged as a critical direction for smart grid cybersecurity due to its capability for early warning and early intervention [11]. Unlike passive detection methods, proactive approaches facilitate early identification and dynamic response to complex attacks behaviors [11]. They offer three key advantages. First

they help prevent physical-level cascading failures through real-time monitoring and behavior analysis [12]. Second, they adapt to evolving threats by employing dynamic policy updates and attack graph-based reasoning [13]. Third, they safeguard data integrity using spatiotemporal anomaly detection [14], [15]. Advanced AI techniques further strengthen proactive defense. Hybrid architectures such as CNN-LSTM models enable multi-source data fusion and early anomaly detection [17], [18]. Spatiotemporal Transformers capture grid patterns via self-attention mechanisms [19] while federated learning enables distributed, privacy-preserving detection in large-scale smart grid environments [20]. However, despite these advancements, practical implementation faces significant challenges, including the need for high-fidelity attack data, real-time adaptation to evolving threats, and balancing detection accuracy with computational efficiency in large-scale grid environments. These limitations highlight the critical need for novel frameworks that can translate theoretical advantages into operational reliability.

To enhance proactive sensing and intervention, deception-based strategies, particularly honeypot techniques, are gaining increasing attention [21]. While mixed H_-/H_∞ optimization effectively balances detection sensitivity and robustness in local physical systems [25], its integration with such proactive deception mechanisms is imperative for achieving resilient defense in large-scale, heterogeneous smart grids. Traditionally, honeypots have proven effective in cybersecurity for attracting attackers and gathering threat intelligence [22], [23], and their growing potential in power systems is becoming increasingly evident as a means to bridge the gap between reactive detection and active defense. By deploying dynamic and scalable decoys, honeypots enable active defenses through attack deception and early intervention. This “deception-for-early-warning” paradigm plays a crucial role in evolving grid security from passive protection to active engagement. Nevertheless, applying honeypots in smart grids introduces unique complexities, such as designing grid-specific lure services, mitigating false alarms induced by legitimate operational variations, and ensuring seamless integration with existing grid control systems. Addressing these challenges is essential to unlock the full potential of honeypot-driven defense, ultimately enhancing grid resilience against sophisticated cyber-physical attacks.

To provide security solutions that are both reliable and adaptive to emerging threats, this paper presents the following contributions. First, a formal framework is established through mathematically defined models of industrial communication sequences, enabling precise characterization of system input–output behaviors and principled identification of potential threats. Second, a multi-layer detection architecture is developed, incorporating data preprocessing, honeypot-driven interaction, dual-path detection, and decision fusion mechanism, along with a gated aggregation strategy for integrating heterogeneous evidence under dynamic conditions. Third, a comprehensive theoretical analysis framework is developed to substantiate the reliability of the proposed detection scheme. The analysis derives a lower bound on feature discriminability, ensuring sufficient separability between normal and malicious behaviors, and establishes an upper bound on the generalization error of the GBDT-based component. Moreover, a

temporal consistency criterion is additionally incorporated to reduce false alarms by enforcing stability across sliding time windows. Finally, the proposed framework is evaluated within operational smart grid environments under diverse and complex attack scenarios. Experimental findings demonstrate that the system achieves high detection accuracy, strong generalization capability, and stable long-term performance.

The rest of this paper is organized as follows. Section II presents the current status and challenges of smart grid security, followed by a detailed description of the system. Section III details a construction of the attack detection network, covering the data preprocessing module, the honeypot interaction layer, the dual-path detection module, and the decision fusion with post-processing mechanisms. Section IV provides a theoretical analysis and feasibility demonstration of the proposed approach. Section V provides experimental evaluations on real-world datasets, demonstrating the effectiveness of the framework in detecting multiple types of cyber-physical attacks. Finally, Section VI concludes the paper and discusses future research directions.

II. PROBLEM FORMULATION

The growing openness and interconnectedness of smart grids enhance operational efficiency and system flexibility but also increase their vulnerability to sophisticated cyber attacks. Current attack detection methods face two major challenges. First, relying on a single detection model makes it difficult to capture both general anomalies, such as data tampering, and specific, targeted attacks, leaving vulnerabilities in the high-dimensional feature space. Second, many AI-based models operate as unobservable systems, providing limited interpretability regarding their decision-making processes. This lack of explainability poses significant risks when deploying such models in critical power infrastructure. Although some studies attempt to fuse data from multiple sources, these approaches are typically reactive, detecting attacks only after they occur and failing to identify early indicators of emerging threats. These limitations collectively lead to a critical gap: the inability to proactively sense and characterize attacks during their initial stages with high confidence and clear reasoning, which is paramount for preventing cascading failures in smart grids.

The smart grid communication environment can be described as a dynamic time-series system. The main idea is to build a clear mathematical connection between what goes into the system and what comes out. Industrial communication sequences are represented as a collection of quadruples:

$$D = \{(t_i, f_i, c_i, s_i)\}_{i=1}^N, \quad (1)$$

where t_i is the timestamp, f_i stands for the protocol field, c_i is the instruction code, and s_i is the device status vector. We use a sliding window approach to break the continuous stream of communication into separate session segments $\{S_j\}_{j=1}^M$. Each session includes Q consecutive interaction events. The core objective of the detection system is to map each session segment S_j to a comprehensive output $\Gamma_j = (y_j, R_j, E_j)$. Here, $y_j \in \{0, 1\}$ is the final attack determination, $R_j \in [0, 1]$ is a fine-grained risk score indicating the severity level, and E_j is a set of evidence supporting the decision. Designing this detection

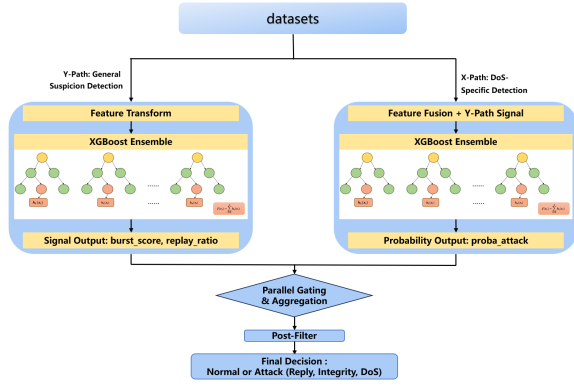


Fig. 1. Architecture of the honeypot-driven proactive detection network for smart grids.

function $\mathcal{F} : D \rightarrow \Gamma$ necessitates balancing multiple critical requirements. First, it must achieve a high detection rate while controlling the false alarm rate below an acceptable threshold α . Second, it must ensure real-time performance such that the average detection delay $E[\tau_{\text{detection}}]$ does not exceed a maximum tolerable time T_{max} . Third, it must provide inherent explainability for every decision, ideally by quantifying the contribution of each feature (e.g., $E_j = \{\phi_i\}_{i=1}^d$, where $\phi_i = \frac{\partial R_j}{\partial f_i}$).

To address the limitations of existing detection methods, this paper proposes a honeypot-driven active detection network. By simulating decoy services to attract and engage attackers, the honeypot generates discriminative behavioral features while providing interpretable real-time evidence for detection. This proactive mechanism shifts defense efforts upstream, capturing attack signatures at the source and overcoming the delayed responses typical of traditional approaches. However, effectively leveraging honeypot interactions requires addressing two key technical challenges, including ensuring the discriminability of extracted features against evolving attacks and designing a robust fusion mechanism that integrates multi-source evidence reliably.

This paper aims to propose an attack detection network framework to meet the aforementioned requirements by construction. Its modular architecture, detailed in the next section, is architected to transform raw communication sequences into trustworthy detection outcomes Γ_j , explicitly tackling the challenges of model generalization, operational stability, and decision transparency outlined in this problem formulation. The next section will describe the detailed structure and implementation of the detection system.

III. ATTACK DETECTION NETWORK FRAMEWORK DESIGN

In this section, we construct the details of the detection network, which is designed to fulfill the requirements of accurate detection, false alarm control, real-time performance, and explainability as outlined in Section II. The system translates the abstract detection function $\mathcal{F} : D \rightarrow \Gamma$ into a practical, layered architecture. As shown in Fig. 1, the system leverages time-series interaction data from grid communications to accurately classify behaviors. It is worth noting that, preprocessing and interaction operate in parallel to minimize detection latency. It computes two key probabilities: a general anomaly score (Path Y) and a DoS-specific probability (Path X). These probabilities, together with honeypot-derived evidence, are

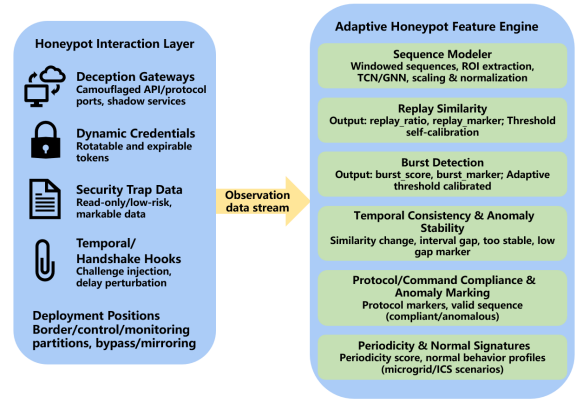


Fig. 2. Feature extraction engine and processing flow in the honeypot-driven detection network.

used to make a final classification into one of four categories: replay attack, integrity attack, DoS attack, or normal operation. The following subsections detail the implementation of each module.

A. Data Preprocessing Module

The data preprocessing module serves as the foundational stage that transforms the raw communication event streams, represented as a collection of quadruples $D = \{(t_i, f_i, c_i, s_i)\}_{i=1}^N$ (1) into a structured and analytically ready format. This module directly operationalizes the mathematical model defined in Section II.

The preprocessing pipeline begins with sessionization, where a configurable Land overlap sliding window is applied to the continuous data stream, breaking it into discrete analyzable sessions $\{S_j\}_{j=1}^M$. This approach balances the need for sufficient contextual information with computational tractability. Each session window then undergoes a comprehensive cleaning procedure. Missing values in temporally dependent fields, such as device status s_i , are imputed via linear interpolation informed by neighboring time points. Meanwhile, categorical variables including protocol fields f_i and instruction codes c_i are transformed through one-hot encoding to yield structured categorical feature representations. Numerical features are subsequently standardized to zero mean and unit variance to prevent model bias towards features with larger scales. Concurrently, a suite of statistical features is extracted from each window, including time-domain characteristics (e.g., mean, variance, and entropy of inter-arrival times) and frequency-domain energy signatures derived via Fast Fourier Transform (FFT). The output is a clean, normalized, and feature-rich representation of each session, forming a reliable base for subsequent detection modules and ensuring the input data quality necessary for achieving the detection goals set forth in Section II.

B. Honeypot Interaction Layer

The honeypot interaction layer acts as the core component of the active defense system. It uses a decoy environment with fake services and simulated systems, as shown in Fig. 2. All interactions occurring within this controlled environment are rigorously recorded, thereby producing a comprehensive

dataset of attacker behaviors. A dedicated feature engine then processes these logs to generate a multi-dimensional evidence vector. The selected features are purposefully designed to capture essential attack characteristics. Replay Similarity measures the proportion of session commands exhibiting strong correspondence to historical attack sequences, quantified via a normalized Hamming distance. Protocol Compliance evaluates adherence to protocol state machines and syntax rules. Burst Intensity calculates the Z-score of request rates to identify flooding patterns. Timing Consistency examines the distribution of inter-packet delays to detect artificial timing patterns typical of replay attacks. Categorical features are transformed using one-hot encoding before GBDT scoring. And honeypots identify integrity violations by detecting deviations from protocol templates

A primary technical challenge is to ensure the extracted features possess sufficient discriminability. This layer addresses this by designing a feature engine that creates multiple evidence types (e.g., replay similarity, protocol compliance) from the interaction data, with the explicit goal of maximizing the separation between attack and normal patterns, thereby satisfying the following condition:

$$D_{total} = \|E[\Phi(S_j)|P_k] - E[\Phi(S_j)|P_0]\|_2 \geq \delta > 0. \quad (2)$$

Each piece of evidence is binarized by comparing it to a set threshold. This practical setup ensures a clear separation between normal and malicious behavior in the features we use, providing the foundational input for the subsequent detection modules.

C. Dual-Path Detection Module

The dual-path detection module employs a collaborative two-model architecture designed to address the limitation of single-model approaches by providing complementary analytical strengths.

The first path, referred to the general anomaly detector (Path Y), is implemented using a gradient boosting decision tree (GBDT) model, specifically the XGBoost algorithm. Its input consists of a fused feature representation that integrates statistical descriptors extracted during preprocessing with binarized threat evidence produced by the honeypot layer. The model is trained to learn complex, non-linear relationships that indicate broad-spectrum anomalies, including replay and integrity attacks. The training process formalizes the model adaptability requirement as an optimization problem:

$$\min_{\theta} E[L(\mathcal{F}_{\theta}(S_j), y_j)] + \lambda R(\theta), \quad (3)$$

where L is the logistic loss function and $R(\theta)$ is an L2 regularization term applied to the weights of the decision trees to control model complexity and prevent overfitting. The output is a calibrated probability score P_Y reflecting the likelihood of a general anomaly.

The second path, designed as the DoS-specific detector (Path X), is a specialized component focusing exclusively on DoS attacks. It takes as input the raw traffic volume time series within the session and is conditioned on the anomaly probability P_Y produced by Path Y, thereby enabling contextual interpretation of high-volume traffic in relation to the overall system anomaly state. Path X is designed to detect the short-term, high-frequency burst patterns that are characteristic of

DoS floods, which may remain inconspicuous in the general feature space captured by Path Y. The output of this module is a dedicated DoS probability P_X .

By combining a general anomaly detector with this specialized DoS-focused pathway, the system achieves complementary detection capabilities, ensuring high sensitivity to both subtle, broad-spectrum anomalies and concentrated, high-volume attacks, thereby significantly enhancing overall robustness and reliability.

D. Decision Fusion and Post-Processing Module

The decision fusion module combines evidence from different sources to make the final decision about an attack, ultimately producing the following output triplet:

$$\Gamma_j = (y_j, R_j, E_j). \quad (4)$$

This module employs a parallel gating structure in which independent triggers operate concurrently on different evidence streams. Specifically, the Replay Trigger is activated when the replay similarity evidence is high AND the general anomaly score P_Y exceeds a threshold. The Integrity Trigger is engaged upon the detection of protocol compliance violations OR abnormal timing consistency, while the DoS Trigger is triggered if the burst intensity is high AND the DoS-specific probability P_X is elevated. This parallel configuration enables the simultaneous detection of co-occurring attacks. In cases where multiple triggers are simultaneously activated, a rule-based conflict resolution mechanism is employed, prioritizing the attack with the highest potential impact, thereby producing a single, deterministic output y_j . Furthermore, to ensure operational reliability and meet the stability requirement, the following measures are implemented:

$$\|\mathcal{F}(S_j + \epsilon) - \mathcal{F}(S_j)\| \leq \beta \|\epsilon\|. \quad (5)$$

This module incorporates a temporal stabilization mechanism, wherein a moving average filter to the risk score R_j over a short, configurable time window. An alarm is triggered only if the smoothed risk score persistently exceeds the predefined threshold for a minimum duration, temporal characteristics are implicitly embedded in the input feature vectors. This straightforward yet effective method achieves exponential suppression of isolated false alarms while maintaining sensitivity to sustained attacks. Due to the shallow architecture of the fusion layer, gradient issues are negligible. Consequently, it ensures the crucial balance between detection sensitivity and operational reliability, which is particularly essential in grid and other high-stakes cyber-physical environments.

This section explains how the honeypot-based detection network is built. The main idea is to turn the theoretical model into a practical solution using a modular design. The data preparation module first organizes raw communication flows into sessions and standardizes them. This creates a foundation for feature extraction. The honeypot interaction layer mimics real services to attract attacks, generating evidence from multiple aspects. The dual-path detection module uses two methods: one provides general anomaly scores from a GBDT model, and the other looks for specific signs of DoS attacks. These two methods work together to improve detection. The decision fusion module then brings all the evidence together

using parallel gating and a temporal stabilization mechanism. Produces a final reliable risk score. The whole system works as an end-to-end process, from raw data to final decision. Its modular design makes it easy to extend or modify.

Remark 1: The four modules form a closely coordinated detection pipeline. Data preprocessing serves as the starting point. It converts raw communication traffic into structured sessions suitable for analysis, delivering a unified input foundation for all following stages. The honeypot interaction layer acts as the core component for active sensing. It simulates a realistic environment to capture first-hand evidence of attacks, supplying critical feature inputs to the detection module. The dual-path detection module leverages the outputs from the previous stages, including structured session data and threat information obtained from the honeypot, to conduct in-depth analysis from two complementary perspectives, namely broad-spectrum anomalies and dedicated DoS attacks, thereby producing initial probability assessments. Finally, the decision fusion module integrates direct evidence from the honeypot with detection results from the dual paths. Through parallel triggering and temporal stability processing, it outputs the final classification and risk score. The entire workflow is data-driven. It not only preserves the functional independence and extensibility of each module, but also ensures that the system operates as an integrated whole, achieving end-to-end detection from raw data to reliable decisions.

With the above architectural design established, the detection network not only accurately identifies attacks but also makes its decisions understandable through feature importance analysis. However, practical deployment in critical infrastructures requires that such a design be grounded in rigorous theoretical justification. To this end, the following section will explore the mathematical foundations of the proposed framework. This includes an analysis of the lower bounds on feature discriminability, the convergence properties of the learning components, and the theoretical mechanism enabling false-alarm suppression. These results collectively substantiate the feasibility and robustness of the system when applied to real-world power-grid environments.

IV. THEORETICAL ANALYSIS AND FEASIBILITY DEMONSTRATION

This section provides a systematic theoretical analysis to ensure the feasibility and robustness of the proposed detection framework. We establish a unified mathematical foundation to verify three key aspects: the usefulness of the feature representations, the learnability and convergence behavior of the model, and the reliability of the overall decision process. Specifically, we will first show that the features created from honeypot interactions, such as replay similarity and protocol compliance, have a clear and measurable difference between attack and normal traffic. We then demonstrate that GBDT algorithm under proper regularization learns very efficiently with exponential convergence. Furthermore, the system's false alarm rate can be reduced exponentially through a window-based consistency check. These results not only provide theoretical support for the deployment of the network's design but also ensure its long-term robustness and explainability.

We first analyze the effectiveness of the feature set.

Theorem 1: (Lower Bound of Feature Discriminability) Consider any feature f_k extracted at the honeypot interaction layer of the industrial communication anomaly detection system. Let P_1 denote the attack distribution and P_0 the normal-operation distribution. Then the expected values of f_k under these two regimes exhibit a strictly positive separation,

$$D_k = |\mathbb{E}_{P_1}[f_k] - \mathbb{E}_{P_0}[f_k]| > 0.$$

Moreover, for the full feature vector $\mathbf{f} = [f_1, f_2, \dots, f_d]^T$, the overall feature discriminability satisfies:

$$D_{\text{total}} = \|\mathbb{E}_{P_1}[\mathbf{f}] - \mathbb{E}_{P_0}[\mathbf{f}]\|_2 \geq \sqrt{\sum_{k=1}^d D_k^2} > 0, \quad (6)$$

which establishes a nontrivial lower bound on the separability of the attack and normal distributions in the feature space.

Proof: We use replay similarity r as an illustrative feature, quantifying the repetition of session interactions relative to historical honeypot responses. Under normal conditions (P_0), the expected value is $\mathbb{E}_{P_0}[r] = p_{\text{normal}}$, while under attack (P_1) it becomes $\mathbb{E}_{P_1}[r] = p_{\text{replay}} + \delta$ with $\delta > 0$. The discriminative power is then:

$$D_r = |\mathbb{E}_{P_1}[r] - \mathbb{E}_{P_0}[r]| = |(p_{\text{replay}} + \delta) - p_{\text{normal}}|. \quad (7)$$

Given that attacks inherently produce more repetition ($p_{\text{replay}} \gg p_{\text{normal}}$) and $\delta > 0$, we have:

$$D_r = \delta + (p_{\text{replay}} - p_{\text{normal}}) > 0. \quad (8)$$

Thus, replay similarity effectively distinguishes attacks. The same rationale applies to other features. For each designed feature f_k , the fundamental pattern difference ensures $D_k > 0$.

For the complete feature vector $\mathbf{f} = [f_1, f_2, \dots, f_d]^T$, let $\boldsymbol{\mu}_1 = \mathbb{E}_{P_1}[\mathbf{f}]$ and $\boldsymbol{\mu}_0 = \mathbb{E}_{P_0}[\mathbf{f}]$. The overall discriminative power is

$$D_{\text{total}} = \|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0\|_2 = \sqrt{\sum_{k=1}^d D_k^2}. \quad (9)$$

Since each $D_k > 0$, we obtain $D_{\text{total}} > 0$, confirming that attack and normal patterns form separable clusters.

For bounded features (e.g., ratios in $[0,1]$), statistical convergence is ensured by Hoeffding's inequality:

$$P(|\hat{f}_k - \mathbb{E}[f_k]| \geq \epsilon) \leq 2 \exp(-2n\epsilon^2). \quad (10)$$

With sufficient samples n , feature estimates converge exponentially to their expectations, guaranteeing the theoretical discriminative power manifests in practice.

The proof establishes both the inherent discriminative capability of features and their statistical estimability, completing the demonstration of Theorem 1. $\square$

This upper bound 10 decays exponentially with sample size n . For any predefined error tolerance $\epsilon > 0$ and confidence level $\delta > 0$, if the sample size satisfies $n > \frac{\log(2/\delta)}{2\epsilon^2}$, we can guarantee with confidence at least $1 - \delta$ that the estimation error $|\hat{f}_k - \mathbb{E}[f_k]| < \epsilon$. This observation provides theoretical guidance for system design: ensuring that sessionization procedure yields communication fragments of sufficient length (n large enough) enables feature estimates to converge to their theoretical expectations with high probability. Consequently, the guaranteed discriminative properties D_k and D_{total} are

reliably reflected in practical observations, thereby enabling effective model training and robust attack–benign separation in real deployments.

We now analyze the convergence properties of the proposed attack detection framework.

Theorem 2 (GBDT Convergence): Let $L(y, F)$ be a convex and twice-differentiable loss function. Under appropriate regularization conditions, the sequence of models $\{F_m\}_{m=1}^M$ generated by the GBDT algorithm exhibits exponential convergence. Specifically, there exists a constant $0 < \rho < 1$, dependent on the condition number of the problem, such that the gap between the empirical risk at iteration m , denoted by $\hat{R}(F_m)$, and the optimal empirical risk $\hat{R}^*$ satisfies the following bound

$$\hat{R}(F_m) - \hat{R}^* \leq K \cdot (1 - \rho)^m,$$

where K is a positive constant. This guarantees that the algorithm converges to the optimal solution at an exponential rate.

Proof: Gradient boosting constructs an additive model that sequentially minimizes a differentiable loss function $L(y, F(\mathbf{x}))$ through iterative weak learner incorporation. At each iteration m , the model undergoes the update $F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + f_m(\mathbf{x})$, where f_m typically represents a decision tree with limited depth, ensuring the weak learner condition.

The fundamental advancement of GBDT lies in its exploitation of both first and second-order gradient information for more informed optimization. This approach provides superior convergence properties compared to first-order methods. The optimization objective in iteration m considers the composite model:

$$\mathcal{L}^{(m)} = \sum_{i=1}^n L(y_i, F_{m-1}(\mathbf{x}_i) + f_m(\mathbf{x}_i)), \quad (11)$$

where n denotes the training sample size. x denotes a d -dimensional feature vector.

When the incremental update f_m is relatively small compared to the current model F_{m-1} , we employ a second-order Taylor expansion around F_{m-1} :

$$L(y_i, F_{m-1} + f_m) \approx L(y_i, F_{m-1}) + g_i f_m(\mathbf{x}_i) + \frac{1}{2} h_i f_m^2(\mathbf{x}_i),$$

where $g_i = \frac{\partial L(y_i, F)}{\partial F} \Big|_{F=F_{m-1}}$ is the first-order gradient and $h_i = \frac{\partial^2 L(y_i, F)}{\partial F^2} \Big|_{F=F_{m-1}}$ is the second-order derivative (i.e., Hessian). This quadratic approximation captures the local curvature of the loss function, leading to more precise updates. Since $L(y_i, F_{m-1})$ remains constant with respect to f_m , the optimization objective is reduced to:

$$\tilde{\mathcal{L}}^{(m)} = \sum_{i=1}^n \left[g_i f_m(\mathbf{x}_i) + \frac{1}{2} h_i f_m^2(\mathbf{x}_i) \right]. \quad (12)$$

For a tree structure partitioning the feature space into J disjoint regions $R_1, \dots, R_J$, we assign a constant weight ω_j to all samples falling into leaf node R_j . The objective function can be regrouped by leaf nodes:

$$\tilde{\mathcal{L}}^{(m)} = \sum_{j=1}^J \left[G_j \omega_j + \frac{1}{2} H_j \omega_j^2 \right],$$

where $G_j = \sum_{i \in R_j} g_i$ and $H_j = \sum_{i \in R_j} h_i$ represent the sum of gradients and Hessians within each leaf. Introducing L2

regularization $\frac{1}{2} \lambda \sum_{j=1}^J \omega_j^2$ with parameter $\lambda \geq 0$ yields the regularized objective:

$$\tilde{\mathcal{L}}_{\text{reg}}^{(m)} = \sum_{j=1}^J \left[G_j \omega_j + \frac{1}{2} (H_j + \lambda) \omega_j^2 \right].$$

This quadratic form admits a closed-form solution by setting the derivative to be zero:

$$\frac{\partial \tilde{\mathcal{L}}_{\text{reg}}^{(m)}}{\partial \omega_j} = G_j + (H_j + \lambda) \omega_j = 0 \quad \Rightarrow \quad \omega_j^* = -\frac{G_j}{H_j + \lambda}.$$

The regularization term λ in the denominator prevents excessively large weights, enhancing generalization.

The convexity of the optimization problem is ensured by the positive definiteness of the Hessian matrix. For the quadratic approximation, the second derivative with respect to ω_j is $H_j = \sum_{i \in R_j} h_i$. Thus, we require $h_i > 0$ for all i to guarantee a unique minimum. For binary classification with logistic loss:

$$L(y, F) = -y \log(\sigma(F)) - (1 - y) \log(1 - \sigma(F)), \quad (13)$$

where $\sigma(F) = 1/(1 + e^{-F})$ is the sigmoid function, and the second derivative is:

$$h = \frac{\partial^2 L}{\partial F^2} = \sigma(F)(1 - \sigma(F)) > 0 \quad \text{for all finite } F,$$

satisfying the convexity condition.

Under appropriate regularity conditions (strong convexity, exact solving), GBDT exhibits linear convergence. After m iterations, the gap between the current loss and the optimal value decreases as $(1 - \rho)^m$, where $0 < \rho < 1$ depends on the problem's condition number. The generalization error bound follows from statistical learning theory:

$$R(F) \leq \hat{R}(F) + \mathcal{R}_n(\mathcal{F}) + \sqrt{\frac{\log(1/\delta)}{2n}}, \quad (14)$$

where $\hat{R}(F)$ is the empirical risk, $\mathcal{R}_n(\mathcal{F})$ is the Rademacher complexity of the hypothesis space $\mathcal{F}$, and the third term is a confidence bound. Regularization techniques effectively control model complexity, ensuring the bound remains tight and preventing overfitting.

The proof thus establishes theoretical convergence guarantees and generalization properties of GBDT, thereby completing the demonstration of Theorem 2. $\square$

Theorem 2 shows the effectiveness and reliability of the GBDT model as the core detector. The analysis provides a theoretical explanation for GBDT's capability to efficiently learn complex nonlinear relationships through an iterative process. The resulting model exhibits strong generalization performance, thereby ensuring the learning capability of the industrial communication anomaly detection system.

Theorem 3 (False Alarm Rate Reduction): The window consistency check mechanism reduces the probability of isolated false alarms from the single-time-point probability p_{FP} to:

$$p_{\text{FP}}^{\text{window}} \approx \binom{W}{k} p_{\text{FP}}^k, \quad (15)$$

where W is the time window size and k is the minimum number of threshold exceedances required for a positive decision, with $k \leq W$. For $p_{\text{FP}} \ll 1$, this mechanism yields an

exponential reduction (of order k) in the resulting false alarm rate.

Proof: The temporal stabilization module continuously monitors risk scores R_t over a sequence of consecutive time segments $t = 1, 2, \dots, W$. This monitoring process forms the basis for reducing transient false alarms while maintaining sensitivity to sustained attacks. For each time segment, we define a Bernoulli random variable X_t that captures the threshold exceedance event:

$$X_t = \begin{cases} 1 & \text{if } R_t > \Theta \text{ (indicating a false alarm occurrence)} \\ 0 & \text{otherwise (normal operation)} \end{cases}$$

where Θ represents the decision threshold for anomaly detection. Under the assumption that false alarm events at different time segments are independent and occur with identical probability, we have $P(X_t = 1) = p_{FP}$ for all t . This independence assumption is reasonable when considering sufficiently spaced time segments where temporal correlations diminish. Consequently, the sequence $\{X_t\}_{t=1}^W$ constitutes independently and identically distributed Bernoulli random variables with parameter p_{FP} .

The window consistency check mechanism implements a robust decision rule where an alarm is triggered only when the cumulative number of threshold exceedances, S_W , within a sliding window of length W reaches or exceeds a predetermined minimum count k , with $k \leq W$. This approach ensures that isolated anomalies do not trigger false alarms while maintaining detection capability for sustained attack patterns. The random variable S_W , representing the total number of exceedances, is defined as the sum of the W independent Bernoulli variables:

$$S_W = \sum_{t=1}^W X_t. \quad (16)$$

By the additive property of independent Bernoulli trials, S_W follows a binomial distribution: $S_W \sim \text{Binomial}(W, p_{FP})$. The probability mass function characterizing this distribution is described as follows:

$$P(S_W = s) = \binom{W}{s} p_{FP}^s (1 - p_{FP})^{W-s}, \quad s = 0, 1, \dots, W.$$

The false alarm probability under the window-based mechanism corresponds to the probability that S_W meets or exceeds the threshold k under normal operating conditions (absence of real attacks):

$$P_{FP}^{\text{window}} = P(S_W \geq k) = \sum_{s=k}^W \binom{W}{s} p_{FP}^s (1 - p_{FP})^{W-s}.$$

In practical industrial anomaly detection systems, stringent reliability requirements dictate that the single-point false alarm rate p_{FP} must be maintained at very low levels (typically $p_{FP} < 0.1$). When $p_{FP} \ll 1$, we can perform an asymptotic analysis of the summation. The dominant contribution comes from the first term ($s = k$) since p_{FP}^s decreases superlinearly with increasing s . The higher-order terms ($s = k + 1, k + 2, \dots, W$) become negligible because p_{FP}^s diminishes by orders

of magnitude relative to p_{FP}^k . This dominance permits the following approximation:

$$P_{FP}^{\text{window}} \approx \binom{W}{k} p_{FP}^k (1 - p_{FP})^{W-k}.$$

Given the small magnitude of p_{FP} , the factor $(1 - p_{FP})^{W-k}$ approaches unity, leading to the simplified expression:

$$P_{FP}^{\text{window}} \approx \binom{W}{k} p_{FP}^k.$$

This result demonstrates that the window consistency check mechanism reduces the false alarm rate exponentially with exponent k when $p_{FP} \ll 1$, providing substantial improvement over single-point detection schemes.

The proof of Theorem 3 is completed. $\square$

Remark 2 (Index reduction effect): The approximation formula $\binom{W}{k} p_{FP}^k$ clearly demonstrates that the false alarm rate after window processing is proportional to the k -th power of the single-point false alarm rate. Since $k \geq 1$ and $p_{FP} < 1$, an exponential reduction in the false alarm rate is achieved whenever $k > 1$. The coefficient $\binom{W}{k}$ represents a binomial coefficient, which typically remains a moderate constant under practical parameter settings.

Notably, while effectively suppressing isolated false alarms, this mechanism has minimal impact on the detection probability of sustained attacks. For a real attack lasting W time segments, the risk score R_t at each segment has a high probability p_D (probability of detection) of exceeding the threshold Θ . In this scenario, S_W also follows a binomial distribution $\text{Binomial}(W, p_D)$. Since p_D is designed to be much larger than p_{FP} , the value of $P(S_W \geq k)$ remains very close to 1, ensuring a high attack recall rate.

Theorem 4 (System Effectiveness Lower Bound): The overall detection probability of the industrial communication anomaly detection system satisfies the following lower bound:

$$P_{\text{system}} \geq 1 - \exp(-n \cdot \min(\text{SNR}_Y, \text{SNR}_X)) - \delta_{\text{cal}}, \quad (17)$$

where n is the sample size used for feature estimation (e.g., number of interactions in a session fragment), SNR_Y and SNR_X represent the signal-to-noise ratios in the feature spaces of Path Y and Path X detectors, respectively, δ_{cal} is the upper bound of error introduced by model score calibration.

Proof: The overall detection process of the system can be decomposed into three sequential stages: feature extraction (F), model inference (M), and decision fusion (D). Successful system operation requires correct execution across all stages. To characterize this behavior, we derive a comprehensive lower bound for the overall success probability by systematically examining the error probability contributions from each individual stage.

Define the following success events:

- E_F : Feature extraction successfully produces discriminative features that distinguish attacks from normal behavior.
- E_M : Model inference correctly maps features to accurate risk scores.
- E_D : Decision fusion properly classifies events based on the synthesized evidence.

The overall system detection probability is given by $P_{\text{system}} = P(E_F \cap E_M \cap E_D)$. Applying the complement form of the union bound from probability theory, we have

$$P_{\text{system}} = 1 - P(\neg E_F \cup \neg E_M \cup \neg E_D) \geq 1 - [P(\neg E_F) + P(\neg E_M) + P(\neg E_D)]. \quad (18)$$

This inequality provides the foundation for establishing a lower bound by analyzing the individual failure probabilities.

Feature extraction fails when the estimated feature vector possesses insufficient signal-to-noise ratio (SNR) to reliably distinguish between attack and normal patterns. From Theorem 1, the theoretical feature discriminability $D_{\text{total}} > 0$ under attack conditions. However, practical estimation from finite samples introduces statistical uncertainty. For bounded features, Hoeffding's inequality bounds the estimation error:

$$P(|\hat{f} - \mathbb{E}[f]| \geq \epsilon) \leq 2 \exp(-2n\epsilon^2).$$

This exponential concentration inequality ensures that with sufficient sample size n , the feature estimation error becomes negligible. The failure probability is consequently bounded by:

$$P(\neg E_F) \leq \exp(-n \cdot \min(\text{SNR}_Y, \text{SNR}_X)), \quad (19)$$

where the min operation ensures robustness regardless of which detection path identifies the attack.

The model inference stage learns the mapping $\mathbf{x} \mapsto s = M(\mathbf{x})$ from features to risk scores. Theorem 2 guarantees that under appropriate regularization, the GBDT algorithm converges with bounded generalization error. Additionally, calibration techniques such as Platt scaling control the calibration error within δ_{cal} , ensuring that the output scores accurately represent true posterior probabilities. The model-specific errors beyond those attributable to poor feature quality are considered higher-order terms, as the primary discriminative capability originates from the feature engineering stage.

Decision fusion integrates scores from both paths through parallel gating and temporal stabilization. The OR-logic parallel gating ensures $P(E_D) \geq \max(P(E_{D_Y}), P(E_{D_X}))$, meaning the system succeeds if either path produces correct detection. Theorem 3 further guarantees that temporal stabilization provides exponential reduction in false alarms while maintaining high detection probability for sustained attacks. Consequently, $P(\neg E_D)$ represents a negligible higher-order contribution compared to the preceding stages.

Integrating the individual error bounds through the union bound framework yields the comprehensive system guarantee:

$$P_{\text{system}} \geq 1 - [\exp(-n \cdot \min(\text{SNR}_Y, \text{SNR}_X)) + \delta_{\text{cal}} + \sigma] \approx 1 - \exp(-n \cdot \min(\text{SNR}_Y, \text{SNR}_X)) - \delta_{\text{cal}},$$

where σ aggregates the higher-order terms from the decision fusion stage. This result provides a rigorous theoretical guarantee for the overall system effectiveness.

The proof of Theorem 4 is completed. $\square$

Remark 3: Through the proofs of Theorems 1-4, this paper develops a complete theoretical framework for the proposed detection network. The derived lower bound on feature discrimination, $D_{\text{total}} > 0$, ensures that behavioral features extracted from honeypot interactions are separable between attack and normal modes. The exponential convergence property of GBDT, together with its generalization

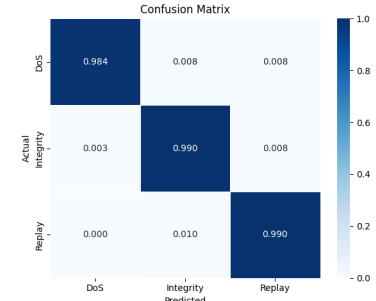


Fig. 3. Attack classification confusion matrix.

error bounds, theoretically supports the learning capability of the model. Moreover, the window-based consistency check mechanism exponentially suppresses the false positive rate, as characterized by $P_{FP}^{\text{window}} \propto p_{FP}^k$, thereby ensuring the system's reliability in practical deployments. Collectively, these results not only validate the soundness of the proposed the network design but also provide explicit mathematical guidance for parameter selection, such as the decision threshold Θ and window size W , reducing reliance on purely empirical tuning.

V. EXPERIMENT

In this section, we evaluate the proposed active defense detection network using real-world datasets. The experimental validation assesses the system's feasibility and robustness through a two-phase approach. Initial model training and performance analysis are conducted on a single-phase representation of a realistic low-voltage grid in Denmark [24]. To examine generalization capability, the trained model is subsequently evaluated on data collected from an IEEE 14-bus microgrid environment, which maintains consistent attack logic while incorporating system noise.

The experimental design strictly follows theoretical guidelines, with emphasis on validating feature effectiveness, model convergence, and decision mechanism rationality. This approach establishes a complete verification chain from theoretical foundation to practical implementation.

A. Model Performance Validation on Danish Low-Voltage Grid Data

This study presents a comprehensive evaluation of the proposed anomaly detection network using real-world communication data collected from an operational low-voltage grid infrastructure in Denmark. The experimental methodology employs rigorous time-series splitting procedures, partitioning the dataset into distinct validation and test sets containing 1,540 and 1,538 samples respectively. Attack scenarios constitute approximately 76% of the total samples, encompassing three critical attack categories: DoS attacks, integrity violations, and replay attacks. This balanced distribution ensures both statistical significance and practical relevance for industrial security applications.

The detection model demonstrates remarkable consistency across both validation and test environments. Attack-specific classification capabilities are quantitatively illustrated through the confusion matrix presented in Fig. 3, which provides detailed accuracy metrics for each attack category. Diagonal

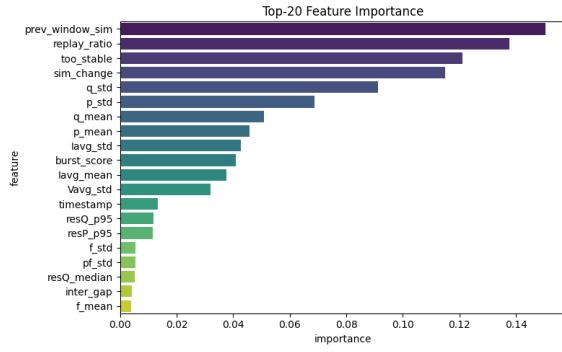


Fig. 4. Top-20 feature importance ranking.

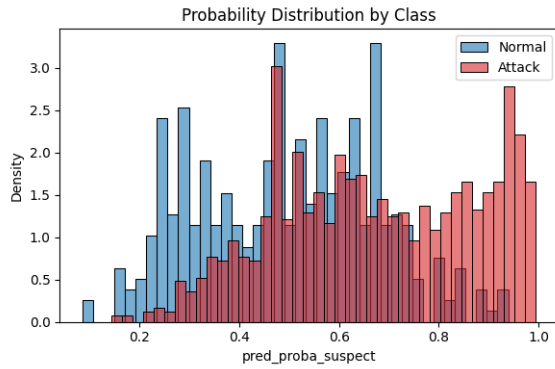


Fig. 5. Class probability distribution comparison.

elements represent correct classification rates, achieving 0.984 for DoS attacks, 0.990 for integrity attacks, and 0.990 for replay attacks. These near-perfect values indicate exceptional recognition accuracy across all attack types. Off-diagonal elements, representing misclassification errors, maintain values below 0.01, confirming minimal false positive rates that satisfy stringent industrial precision requirements. On the independent test set, all three attack categories maintain detection rates exceeding 98.4%, with DoS attacks at 98.4% while integrity and replay attacks approach 99.0% accuracy. This consistently high performance across diverse attack scenarios validates the effectiveness of the parallel gating mechanism in complex operational environments.

Feature importance analysis, depicted in Fig. 4, offers valuable insights into the underlying operation of the detection mechanism. The three most influential features, namely previous window similarity (0.15), replay ratio (0.14), and excessive stability (0.12), are all originate from the honeypot interaction layer and exhibit substantially stronger discriminative capability than the remaining features. This concentration of honeypot-derived features provides empirical validation for the theoretical principle of a lower bound on feature discriminability established earlier in this work. In addition, the top four features together contribute more than 50% of the overall feature importance, demonstrating the efficiency of the proposed feature engineering approach, which achieves reliable decision-making while keeping computational complexity low.

Probability distribution analysis, illustrated in Fig. 5, reveals distinct separation patterns between normal and attack classifi-

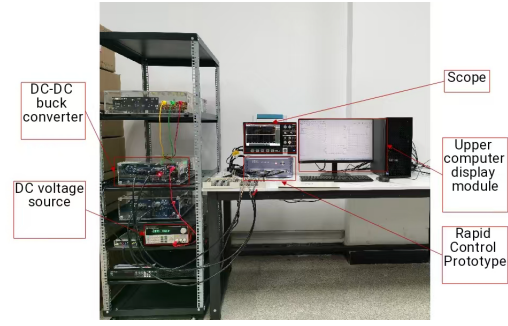


Fig. 6. Hardware-in-the-loop security testbed for IEEE 14-Bus microgrid.

cations. Attack samples demonstrate significant concentration in the high-probability region (0.8-1.0), while normal samples exhibit substantially lower densities within this critical range. This clear probabilistic separation establishes a reliable foundation for threshold configuration and enables precise alert generation in high-risk operational scenarios. The well-defined boundary between classes further confirms the model's calibration quality and its suitability for practical deployment.

Generalization capability assessment through cross-dataset performance comparison reveals exceptional stability. Key detection metrics maintain consistency with less than 0.5% degradation in detection rates across all three attack categories between validation and test environments. This minimal performance variation confirms the temporal stability of learned attack patterns and the system's inherent adaptability to natural grid communication variations. The results collectively demonstrate the practical viability and operational value of the detection system in real-world low-voltage grid environments, establishing it as a reliable security solution for industrial control systems requiring robust anomaly detection capabilities.

B. Model Generalization Validation on IEEE 14-Bus Microgrid Data

The experimental platform for data acquisition is depicted in Fig. 6, comprising core components including a DC-DC buck converter, DC voltage source, oscilloscope, and rapid control prototyping hardware. This setup provides reliable experimental conditions for collecting communication data with system noise in an IEEE 14-bus microgrid environment. The detection model, previously trained on Danish low-voltage grid data, is directly applied to this new environment to evaluate the practical performance of the honeypot-based active defense detection network in cross-system deployment scenarios.

Experimental results demonstrate excellent generalization capability and robustness. On the validation set, the model achieved an accuracy of 0.989, with perfect normal sample recall (1.0) and attack sample recall of 0.987. When evaluated on the test set containing mild system noise, the model maintained high performance with an accuracy of 0.974, stable normal sample recall (1.0), and attack sample recall of 0.968. Although test set performance showed approximately 1.8 percentage points degradation due to noise interference, this reduction remains within acceptable limits, indicating overall stability.

Analysis of specific attack categories reveals strong detection capabilities across different attack patterns. DoS attacks achieved the highest detection rate (0.984), while replay and

integrity attacks maintained detection rates of 0.959 and 0.955, respectively. Notably, DoS attacks demonstrated the strongest noise immunity under current threshold settings. Both replay and integrity attacks maintained detection rates above 95% under mild noise conditions, confirming the significant discriminative power and stability of attack features extracted by the honeypot-based active defense mechanism.

Validation on the practical microgrid experimental platform further demonstrates the advantages of the proposed detection network, including strong generalization ability, modular design that allows integration of new specialized detectors, high tolerance to system noise, and a favorable balance between accuracy and robustness. These results indicate that the proposed approach provides a reliable and effective technical solution for power system security protection.

VI. CONCLUSION

This paper addresses the challenge of industrial communication anomaly detection in smart grids by proposing a honeypot-based active defense framework. A hierarchical detection architecture is designed that integrates feature extraction, dual-path modeling, and decision fusion to accurately identify multiple attack types while maintaining operational reliability. Theoretical analysis establishes fundamental performance guarantees, including lower bounds on feature discriminability, model convergence properties, and system-wide effectiveness thresholds. Experimental validation using real-world power grid data demonstrates stable performance, achieving detection accuracies exceeding 98.4% for DoS, integrity, and replay attacks, with only degradation under noisy conditions. Furthermore, the proposed implementation translates theoretical insights into practical detection modules, incorporating adaptive threshold calibration and evidence-based decision mechanisms. Overall, the framework provides a mathematically grounded and practically deployable solution for power grid cybersecurity, achieving high detection accuracy and robustness without compromising operational stability.

Future work will focus on developing detection systems that are more robust and adaptive. A primary focus is on deeply integrating the physical laws of power grids as hard constraints into the models. This allows the system to internalize these principles, which naturally rules out unrealistic attack patterns and makes its decisions more interpretable and trustworthy. Another important direction involves creating online adaptive mechanisms. Since the grid topology and operating conditions constantly change, the system must dynamically adjust. Using real-time feedback on its performance such as false alarms and novel attack samples, which can continuously refine its detection thresholds and parameters. The goal is to shift from static defense to a self-evolving one, ultimately forming an active defense framework that seamlessly blends physics-based knowledge with data-driven learning for ongoing adaptation.

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